Titanite

△ **Classic** wedge-shaped titanite crystals on a rock matrix

Formerly known as sphene, (the Greek for "wedge"), titanite is a common mineral in many igneous rocks and in metamorphic rocks such as gneiss and schist. It occurs as translucent or transparent crystals. It is found in numerous locations, and can occur as reddish brown, grey, red, yellow or green monoclinic crystals. Its "fiery" colour results from its high level of dispersion and high refraction index. In addition to its use as gems, it is also a source of titanium dioxide, which is used in pigments.

Specification

Chemical name Calcium titanium silicate | **Formula** $CaTiSiO_5$ | **Colours** Yellow, green, brown, black, pink, blue | **Structure** Monoclinic | **Hardness** 5–5.5 | **SG** 3.5–3.6 | **RI** 1.84–2.11 | **Lustre** Vitrous to greasy | **Streak** White | **Locations** Europe, Madagascar, Canada, USA, Brazil, Russia, Pakistan

Crystal on rock groundmass | **Rough** | Lozenge-shaped titanite crystals cover the top of the rock matrix in this superb collector's specimen.

Faceted oval | **Cut** | This oval brilliant-cut titanite stone has been expertly faceted. Its naturally dark yellow colouring gives its cut a dense appearance.

Rectangular titanite | **Colour variety** | This rectangular, step-cut gemstone has a lower iron content, resulting in the clear, yellow-green colour seen here.

Butterfly with titanite | **Set** | Madagascar is the source of the 11 fine-quality, brilliant titanite gems that provide the sparkle for this 18-karat gold butterfly brooch with sapphire eyes.